Spatial and Temporal Characterization of Cardiac Neural Crest and Schwann Progenitor Cell Populations During Human Heart Development

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Abstract

Cardiac neural crest cells (cNNCs) and Schwann progenitor cells (SPCs) are important contributors to human heart development and cardiac innervation. Although both populations are associated with neural crest-derived developmental processes, the extent to which they represent continuous developmental lineage or functionally distinct cell populations remains unclear. In this study, we integrated single-cell RNA sequencing (scRNA-seq), spatial transcriptomics (ST), and in situ sequencing (ISS) data from the developing human heart to investigate the transcriptional, temporal, and spatial relationships between cNCC-like and SPC-like populations.

scRNA-seq data were used to identify transcriptional signatures and marker genes associated with cNCC-like and SPC-like states. These signatures were then applied to ST datasets to generate spatial module scores and characterize temporal abundance patterns across developmental stages. In parallel, ISS data were processed using image-based cell segmentation and transcript aggregation to assign cell identities at near single-cell spatial resolution. Spatial neighborhood enrichment analysis was then performed to evaluate co-localization patterns between predicted cell populations.

Our analyses revealed partial transcriptional overlap between cNCC-like and SPC-like populations, accompanied by distinct temporal and spatial organization. SPC-associated signatures showed transient enrichment during mid-developmental stages, while neighborhood enrichment analysis demonstrated preferential clustering within the same predicted cell types. Together, these findings support a model of partial developmental continuity accompanied by increasing spatial and functional specialization during heart development.

1 Introduction

The development of the human heart requires the coordinated interaction of multiple transient progenitor cell populations that contribute to cardia structure, innervation, and vascular organization. Among these populations, cardiac neural crest cells (cNCCs) play essential roles in outflow tract septation, autonomic innervation, and cardiac morphogenesis. Schwann progenitor cells (SPCs), which are associated with peripheral glial development, have also been implicated in cardiac neural crest development and may share developmental relationships with neural crest-derived populations. However, the extent to which cNCCs and SPCs represent a continuous developmental trajectory or distinct functional cell states remains incompletely understood.

Advances in single-cell and spatial transcriptomic technologies have enabled increasingly detailed characterization of cellular organization during embryonic development. Single-cell RNA sequencing (scRNA-seq) provides high-resolution transcriptional profiling of individual cells, allowing identification of cellular subpopulations and developmental programs. However, dissociation-based sequencing approaches remove cells from their native tissue context and therefore lose important spatial information. Spatial transcriptomics (ST) and in situ sequencing (ISS) address this limitation by preserving spatial organization while measuring gene expression directly within tissue sections.

Previous work by Måns Asp and colleagues generated a comprehensive spatiotemporal atlas of the developing human heart using integrated scRNA-seq, ST, and ISS technologies. Their study identified distinct neural crest-associated populations within the developing human heart, including cardiac neural crest and Schwann progenitor populations, but the developmental relationships between these groups was not fully resolved.

In this project, we integrated scRNA-seq, ST, and ISS datasets from the developing human heart to investigate the molecular, temporal, and spatial relationships between cNCC-like and SPC-like populations. scRNA-seq data were used to derive transcriptional signatures and marker genes that were associated with each population. These signatures were then applied to ST datasets to evaluate developmental stage-specific abundance patterns and spatial localization. In parallel, ISS datasets were analyzed at near single-cell resolution to characterize spatial neighborhoods and local cellular organization.

By combining the transcriptional, temporal and spatial analyses, we aimed to evaluate whether cNNC-like and SPC-like populations are better explained by a developmental continuity model, characterized by partial overlap and progressive transition, or by a functional separation model, characterized by distinct transcriptional identities and segregated spatial niches.

2 Data Preparation and Methods

All datasets used in this project were obtained from the developmental human heart atlas published by Måns Asp and colleagues. The Study provides integrated single-cell RNA sequencing (scRNA-seq), spatial transcriptomics (ST), and in situ sequencing (ISS) data spanning multiple stages of early human heart development. Developmental stages analyzed in this project included 4.5, 6.5, and 9.5 post-conception weeks (PCW)

3 Experimental Design

This project uses the Asp et al. (2019) human developmental heart atlas to investigate the transcriptional, spatial, and temporal relationship between cardiac neural crest cells (cNCCs) and Schwann progenitor cells (SPCs). First, scRNA-seq data were used as a reference to refine the mixed "cardiac neural crest cells & Schwann progenitor cells" cluster. Marker-based scoring was applied using cNCC-associated markers and SPC-associated markers, followed by PCA visualization and differential expression analysis to identify subtype-enriched gene signatures.

Next, spatial transcriptomics (ST) data were analyzed to estimate the spatial abundance of cNCC-like and SPC-like populations across tissue spots. ST count data were library-size normalized and log-transformed. Because the ST matrix used Ensembl gene IDs while the scRNA-seq reference used gene symbols, an Ensembl-to-gene-symbol mapping step was added before integration. DEG-derived cNCC-like and SPC-like signatures were then used to compute per-spot abundance scores. These scores were visualized using spatial maps, PCA/UMAP plots, and stage-wise abundance plots across developmental weeks 5, 6, and 9.

In parallel, in situ sequencing (ISS) data were used as a higher-resolution spatial validation track. ISS transcript coordinates were assigned to segmented cells, cell-by-gene matrices were generated, and cells were classified as cNCC, SPC, or other based on marker scores. Spatial neighborhood and co-occurrence analyses were then used to assess whether cNCCs and SPCs occupy overlapping or distinct spatial niches.

Together, this design integrates scRNA-seq, ST, and ISS to test whether cNCCs and SPCs represent a transcriptionally related developmental continuum or spatially and functionally distinct populations during human heart development.

4 Result

4.1 Spatial Transcriptomic

To investigate the relationship between cardiac neural crest cell (cNCCs) and Schwann progenitor cell (SPC), cells from the selected developmental cluster were classified into two putative subtypes using marker-based scoring derived from known lineage-associated genes (Asp et al. 2019). The resulting subtype assignments are summarized in this file (predicted_cnc_schwann_subtypes_metadata.csv).
Cells exhibiting elevated expression of canonical neural crest markers were labeled as Cardiac neural crest-like, where cells enriched for Schwann markers were labeled as Schwann progenitor-like. This produced a refined subtype annotation framework for downstream molecular and spatial analyses.

The transcriptomic distribution between the two predicted subtypes was visualized using PCA. From Figure 1, the plot demonstrated a clear separation between cNCCs-like cells and SPC-like cells. Despite originating from the same parent cluster, the two groups occupied distinct regions of transcriptional space, indicating that they represent biologically different molecular states rather than random variation within a single homogeneous population. This separation supports the effectiveness of the marker-based scoring strategy and suggests that the selected markers successfully captured subtype specific transcriptional programs.

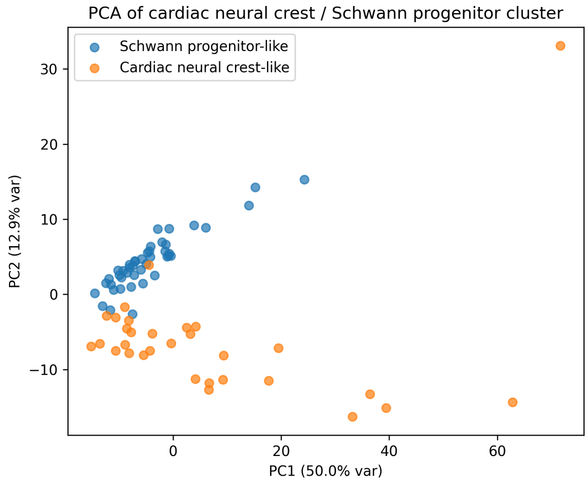

Figure 1: Spectrogram with energy and pitch contours

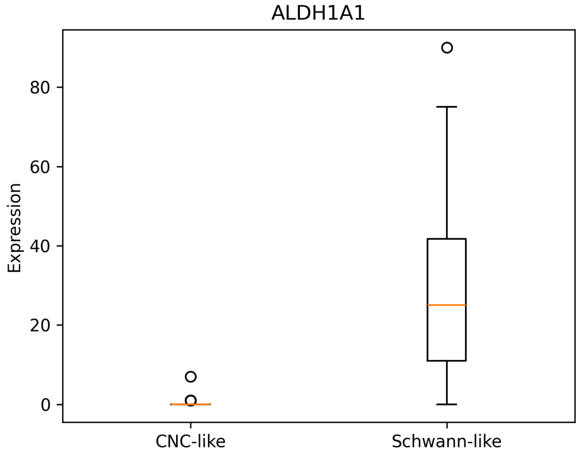

Figure 3: ALDH1A1 box plot

To validate the subtype assignment, expression patterns of key lineage-associated markers were examined.

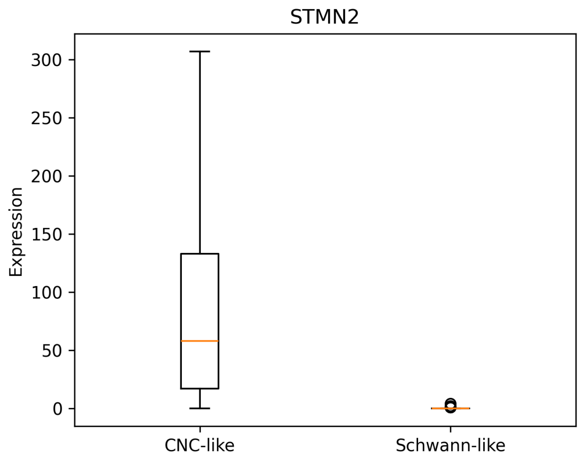

Figure 2: STMN2 boxplot

Figure 2 showed that STMN2 expression was significantly elevated in cNCC-like cells and the expression was substantially reduced in SPC-like cells. Since STMN2 is associated with immature neural and migratory neural crest-related states, its enrichment in the CNC-like population supports the interpretation that these cells retain earlier neural crest characteristics.

Figure 3 showed that ALDH1A1 expression was highly enriched in SPC-like cells and the expression was minimal in cNCC-like cells. ALDH1A1 has been associated with glial differentiation and Schwann lineage maturation. Its strong enrichment in Schwann-like population provides additional validation that the classification successfully identified a distinct glial-associated cellular state. Together these marker analyses confirm that the subtype classification strategy accurately partitioned cells according to biologically meaningful transcriptional identities.

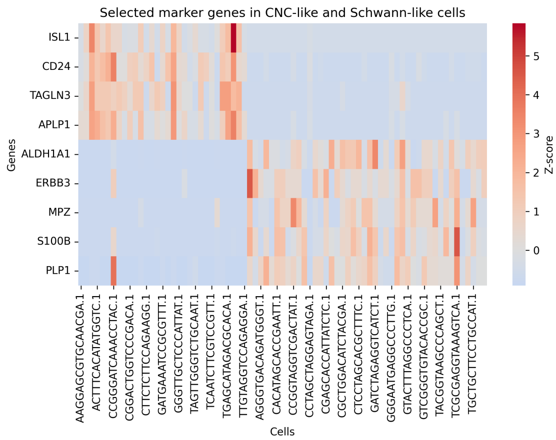

Figure 4: Selected marker genes in CNC-line and Schwann like

The subtype-specific transcriptional programs were further examined. Figure 4 revealed distinct molecular signatures for each population cNCC-like cells, and SPC-like cells. cNCC-like's enriched gene included ISL1, CS24, and STMN2 which are associated with neural crest identity, developmental plasticity, and migratory and progenitor-like state. SPC-like 's enriched genes included ALDHA1A, ERBB3, S100B, and MPZ, which are characteristic of Schwann lineage commitment, glial differentiation, and peripheral nervous system maturation. The overall expression patterns indicate that the two populations possess distinct but biologically related transcription programs consistent with different stages along a neural crest-to-glial developmental axis.

Differential expression analysis results are summarized in cnc_vs_schwann_like_DEG_results.csv. The analysis identifies multiple significantly differentially expressed genes between the two subtypes, further supporting their molecular distinction. There are two notable findings include, VIM which was significantly elevated in SPC-like population and HSPB2 which also showed subtype-specific enrichment. The increased expression of VIM in the SPC-like cells may indicate the enhancement of mesenchymal characteristics, the increased cellular maturation and the progression toward a glial-supportive phenotype. In contrast, the CNC-like cells retained stronger expression of genes associated with earlier developmental and progenitor-like neural crest states.

Collectively, the PCS separation, marker validation, heatmap analysis, and differential expression results provide strong evidence that the original cluster contains at least two transcriptionally distinct populations, cNCC-like and SPC-like. The observed expression patterns suggest a potential developmental progression in which neural crest-like cells transition toward a more glial-associated Schwann progenitor state. However, the clear transcriptomic separation between the two groups also indicates that they maintain distinct molecular identities rather than forming a completely homogeneous continuum.

4.2 In Situ Sequence

The pipeline assigns single-cell spatial identities to cells in situ, using ISS gene expression and classified each cell as one of three types: cNCC, SPC or other. Classification uses marker gene scoring, ISL1 and STMN2 for cNCC; ALDH1A1 for SPC and then runs spatial statistics to understand how these populations organize in tissue.

4.2.1 4.5 PCW

From figure 5, on panel A, only a few cNCC are detected with a scattering of other cells across the section. SPC are absent. Panel B showed a tight focal cluster of ISL and STMN2 cells in the upper portion of the sections. This is spatially consistent with neural crest migration routes but the count is very low. Only one ALDH1A1 is detected and the SPC is essentially empty. This is biologically plausible since at 4.5PCW, second heart field progenitors may not yet strongly express ALDH1A1. Panel D shows only a single cNCC cell visible and no SPC which confirms the minimal signal at this stage. Panel E showed only the single detected cNCC cell that shows a high score. Panel F showed a gradient of SPC scores is visibles along what appears to be a curved anatomical structure. While no cells are classified as SPC, there is a weak score signal along this trajectory which potentially reflects low level ALDH1A1 expression in progenitor-adjacent cells.

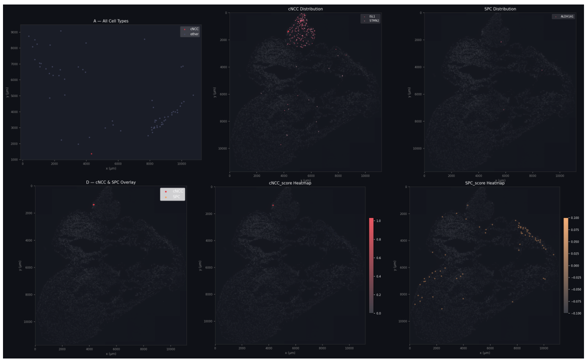

Figure 5: Spatial Panel

From figure 6, two panels are shown. The left panel, the “other” line holds flat at 1 while the cNCC line stays near 0, indicating that given a cNCC anchor cell, the probability of finding any cell type in the neighborhood does not rise above chance. This reflects the near-absence of cNCC cells. The right panel, when anchored on an “other” cell, the cNCC co-occurrence rises with distance. This is a very subtle positive trend, likely a statistical artifact of having only 1-2 cNCC cells at large distances; the one cNCC cell becomes detectable from most points in the tissue. Overall, co-occurrence is uninformative at 4.5 PCW due to insufficient cell numbers. No biologically meaningful spatial clustering can be inferred.

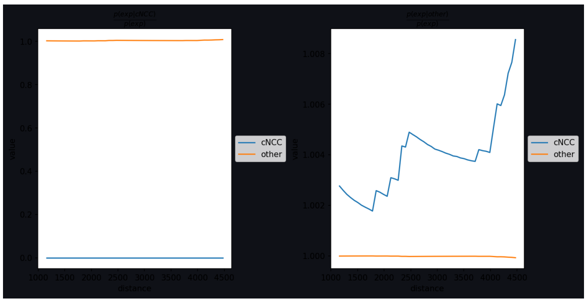

Figure 6: PCW4.5 co-occurrence

From figure 7, the cNCC-cNCC block is

grey.The other-other block is yellow indicating strong self-association of the “other” population, these cells clutter with each other spatially. The other-cNCC is purple, indicating that cNCC and other cells are spatially depleted from each other’s neighborhoods. This is primarily a reflection of the very small cNCC sample rather than a true biological boundary.

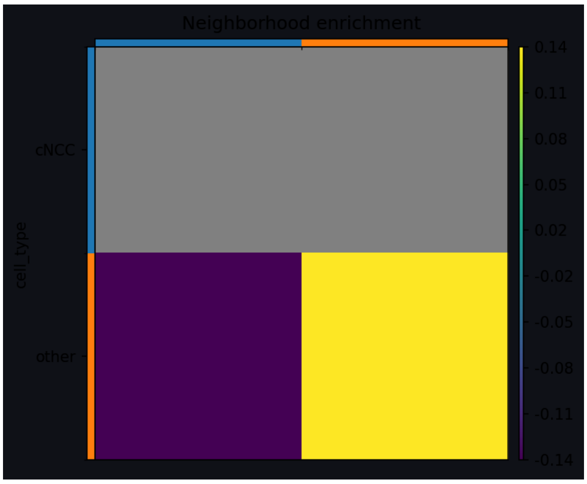

Figure 7: PCW4.5 neighborhood enrichment

4.2.2 6.5 PCW

From figure 8, on panel A, both cNCC and SPC are presented, distributed across the section. The “other” cells fill the full tissue extent. The density of detected cells has increased dramatically compared to 4.5 PCW. Panel B shows cNCC cells concentrated in the upper left region of the section, likely corresponding to the outflow tract/conotruncal region and and pharyngeal mesenchyme, precisely where cardiac neural crest cells are expected to migrate and colonize. A distinct focal hotspot of ISL1 and STMN2 cells is visible. On SPC distribution, ALDH1A1 and SPC cells show a complementary but overlapping distribution to cNCC. They are present across the superior portion but with some extension into the ventricular region. The SPC signal is diffuse rather than tightly focal. Panel, D show the overlay reveals spatial colocalization of cNCC and SPC in the superior tract region, with individual cells of both types intermingled. This is consistent with the transitional zone where neural crest and second heart field progenitors interact during cardiac morphogenesis. Panel E shows strong, broad cNCC scoring across nearly the entire upper half of the tissue section, including the outflow tract and part of the atria. Scores extend well beyond the individually called cNCC cells, suggesting many cells have intermediate marker expression. This is a rich signal not seen at 4.5 PCW. Panel F, shows SPC score is similarly broad and strong across the superior section, following a similar spatial domain as the cNCC score but not identical. The SPC signal is prominent in the region corresponding to the anterior heart wall.

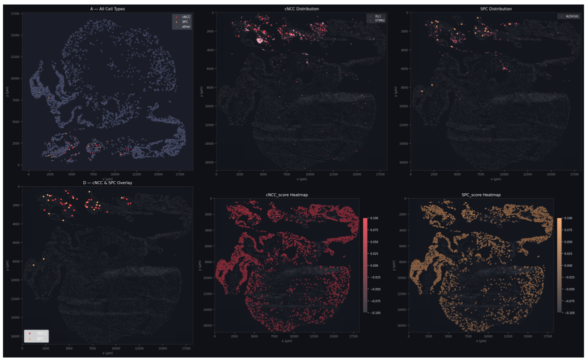

Figure 8: PCW 6.5 Spatial Panel

From figure 9, the left panel, the SPC anchored, both SPC and cNCC show strong enrichment at short distance, decaying rapidly with distance while other cells stay flat. This indicates that near an SPC cell, there is more likely to find another SPC or cNCC cell than by chance, showing a strong colocalization signal. The cNCC-anchored (middle panel), shows both SPC and cNCC are highly enriched at short range and falling, while the other cell is flat. This confirms reciprocal spatial clustering of cNCC and SPC. The Other-anchored (right), show the SPC and cNCC are depleted near the other cells while other cell are flat. This confirms that the cNCC/SPC cluster is spatially segregated from the broader "other" cell mass.

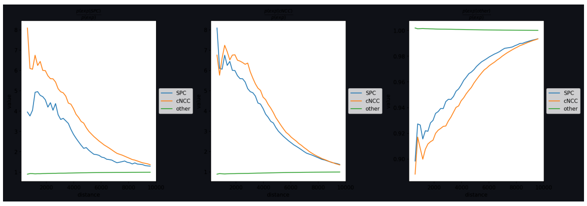

Figure 9: PCW 6.5 co occurrence

Figure 10, shows several key features. The SPC-SPC (teal) show SPC cells are enriched near other SPC. The cNCC-cNCC (yellow) shows cNCC cells strongly cluster together. SPC-cNCC cross (yellow) shows cNCC and SPC are strongly co-enriched in each other’s neighborhoods. The cNCC-other (purple) shows cNCC are significantly depleted from the other neighborhoods and vice versa. The SPC-ther (blue) shows SPC also tends away from the other cells.

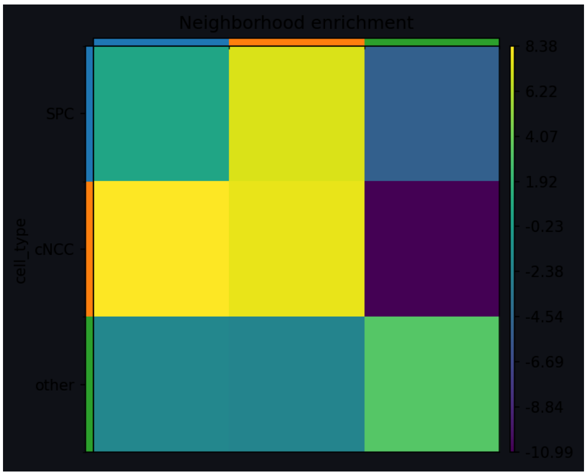

Figure 10: PCW 6.5 neighborhood enrichment

4.2.3 9.5 PCW

From figure 11, Panel A, shows both cNCC and SPC are present but more scattered across the section compared to 6.5 PCW. Some focal groupings remain (upper-left region), but cNCC cells are also seen at multiple distinct locations including the ventricular walls and in the lower section. "Other" cells fill the full tissue extent densely. Panel B, shows a focal cluster of high ISL1, STMN2, and cNCC cells remains in the upper-left/outflow tract region, but several individual cNCC cells are scattered elsewhere. This reflects the ongoing dispersal of neural crest derivatives as they contribute multiple cardiac compartments. Panel C shows SPC cells (ALDH1A1+) are similarly scattered, with some concentration in the upper-anterior region. The distribution is broader and less focal than at 6.5 PCW, consistent with dispersal of second heart field derivatives into ventricular myocardium. Panel D shows the overlay shows spatial separation emerging cNCC and SPC are no longer as tightly co-localized as at 6.5 PCW. They occupy overlapping but distinct zones. Some clustering remains near the outflow tract, but the tight intermingling seen at 6.5 PCW has dissipated. Panel E shows strong cNCC scoring in the upper portion, spanning the outflow tract and anterior wall, but weaker and patchier than at 6.5 PCW in the lower regions. More cells score in the intermediate range, reflecting a broader but diluted cNCC signature. Panel F shows SPC score is broadly distributed across the upper portion of the section, similar spatial domain as cNCC but extending slightly further into the ventricular walls. The signal is diffuse rather than concentrated.

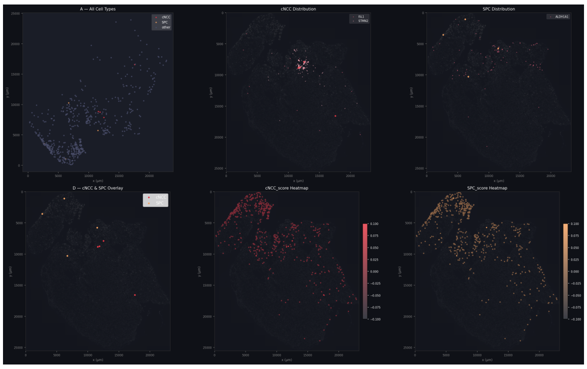

Figure 11: PCW 6.5 Spatial Panel

From figure 12, the SPC-anchored (left), cNCC shows only modest enrichment near SPC. Unlike 6.5PCW, the enrichment is weak and noisy. SPC self-enrichment oscillated while the other stayed flat. The cNCC-anchored (middle), the cNCC shows extremely strong self-clustering. This reflects the focal hotspot of cNCC cells in the outflow tract, when anchored on one cNCC, there is almost certain to find another nearby. The SPC and others show near-zero enrichment from cNCC. The other-cell-anchored (right), SPC and cNCC are mildly depleted from Other neighborhoods. Other cells are no longer as strongly excluded from cNCC/SPC zones as at 6.5 PCW.

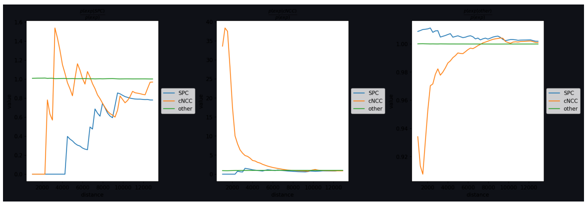

Figure 12: PCW 6.5 Co-occurrence

From figure 13, The enrichment values are substantially reduced compared to 6.5 PCW. The cNCC-cNCC (yellow), still self-enriched but range is narrower than 6.5 PCW. The cNCC-SPC (teal), the strong positive co-enrichment of 6.5 PCW has disappeared. cNCC and SPC are now spatially neutral relative to each other. The cNCC-other (purple), cNCC still somewhat depleted from Other neighborhoods but less extremely so than at 6.5 PCW. The SPC-SPC and SPC-Other (teal), SPC has largely dispersed into the general tissue.

Asp et al. 2019 used Spatial Transcriptomics to map gene expression across the developing hu-

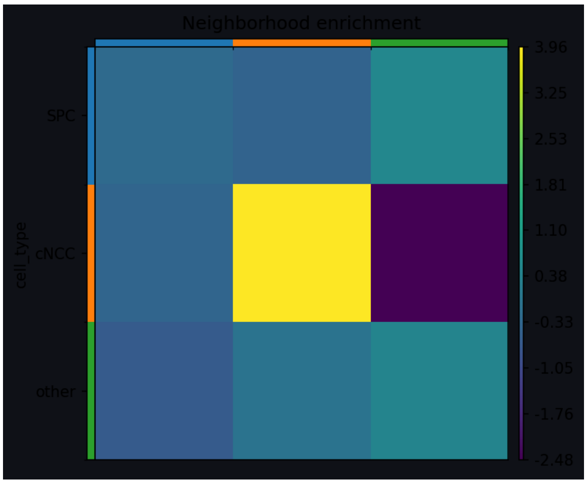

Figure 13: PCW 6.5 neighborhood enrichment

man heart at multiple stages including 4.5–9 PCW. This ISS pipeline uses single-cell resolution ISS to assign individual cell identities using a targeted marker panel. The result from the ISS pipeline confirm several finding from Asp 2019, including, cNCC localize to outflow tract, SPC (ALDH1A1) present in cardiac progenitor zones, cNCC and SPC co-localize during outflow tract development, early stages (<5 PCW) show sparse signal and later stages show dispersal of progenitors into multiple compartments. However, there is also a difference in resolution and analytical capability between the primary paper and this pipeline. The primary advancement of this new pipeline is the transition from qualitative "spot-level" observation to quantitative "single-cell" statistics. This allows for a much more rigorous mathematical understanding of how cells are organized and how they interact within their microenvironment.

5 Cross Modality Validation

The cross-validation pipeline overlays ISS onto ST coordinate space and tests whether ST deconvolution scores and ISS cell counts agree spatially. A spot with a high ST cNCC deconvolution score should also contain a high density of ISS-assigned cNCC cells in the same tissue region.

5.1 4.5 PCW

From figure 14, ISS data is extremely sparse, with only 1 or 2 cells detected across the entire tissue. This results in a NaN correlation, as the near-zero variance in ISS data makes mathematical correlation impossible.

From figure 15, and the correlation metrics, con-

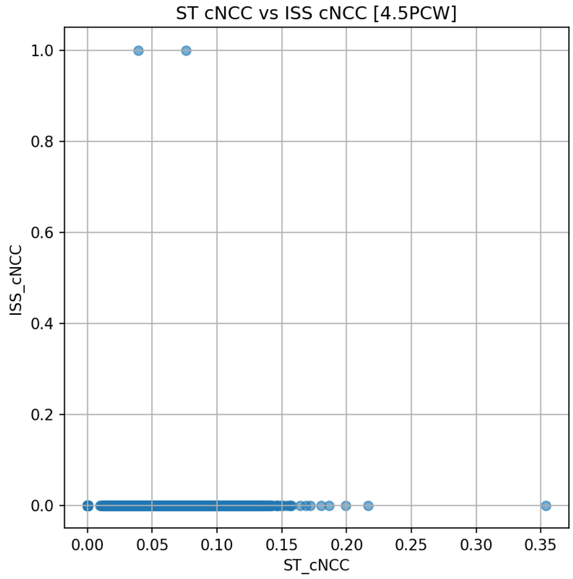

Figure 14: PCW 4.5 cNCC concordance

firm the mismatch is absolute. ISS detected zero SPC cells, while ST deconvolution assigned scores up to 0.32 across multiple spots. The correlation (r = -0.006) confirms there is no relationship between the two modalities at this stage.

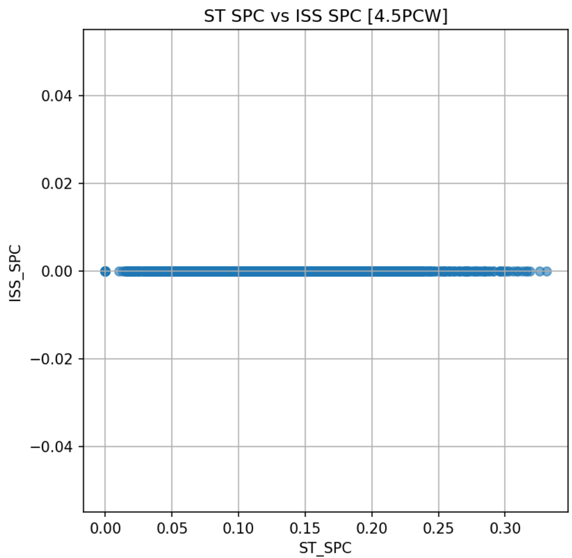

Figure 15: PCW 4.5 SPC concordance

The failure to cross-validate is not a computational error, but a biological and technical signal.ST uses the whole transcriptome to infer cell types. Even if specific markers like ALDH1A1 are lowly expressed, ST can still detect the cell type through thousands of other low-signal genes.ISS relies on a targeted 3-gene panel. If the specific marker genes are below the ISS detection threshold at 4.5 PCW, the modality returns a zero, even if the cell type is transcriptomically present.

5.2 6.5 PCW

From figure 16 and the correlation metrics, ISS detects a clear gradient of cells , while ST provides continuous deconvolution scores. This is the only statistically significant result in the study ($p < 0.001$), but the correlation is negative ($r = -0.06$). Spots identified as high-density by ST tend to have fewer cells according to ISS. This suggests a systematic disagreement rather than random noise, potentially caused by coordinate registration offsets or regional sensitivity differences between the two methods.

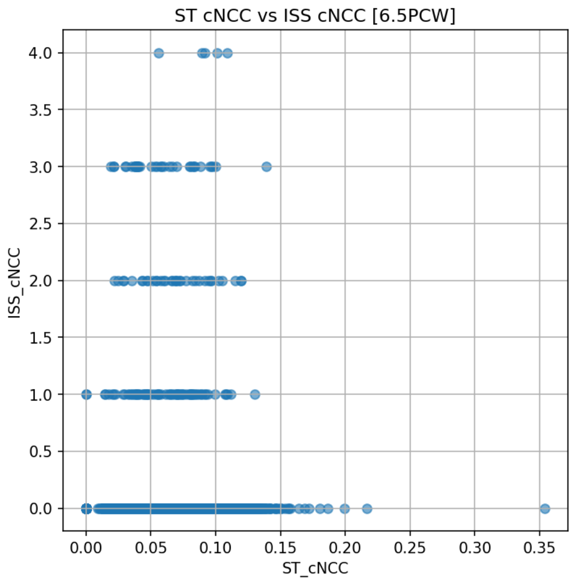

Figure 16: PCW 6.5 cNCC concordance

From figure 17 and the correlation metrics, ISS shows three distinct bands (0–3 cells), and ST shows a broad range of scores. The correlation is effectively zero ($r = -0.002$, $p = 0.924$). An ISS-detected SPC is just as likely to appear in a low-score ST spot as a high-score one. There is no spatial relationship between where each modality "sees" these cells.

From figure 18 and the correlation metrics, ISS detects 0–3 cells per spot, while ST scores remain capped at 0.35. Correlations are effectively zero ($r = -0.008$), indicating no concordance. Spots with the highest ISS cell counts (2–3 cells) actually cluster at lower ST scores (below 0.10). This behavior suggests that where ISS sees the most cells, the ST deconvolution sees the weakest signal.

From figure 19 and the correlation metrics, ISS

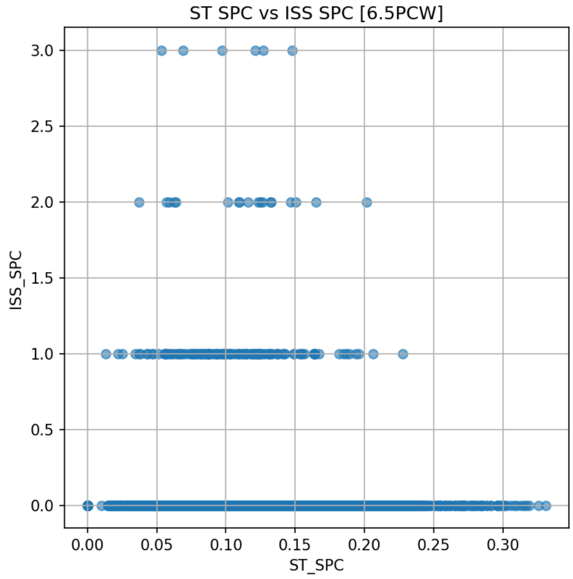

Figure 17: PCW 6.5 SPC concordance

detection is limited to only 0 or 1 cell per spot, a significant drop from the 6.5 PCW stage. Correlation is non-existent ($r = -0.005$). The few spots containing an ISS-detected SPC are scattered randomly across the range of ST scores (0.03 to 0.23). The drop in ISS counts aligns with the spatial dispersal of SPC cells into the broader tissue at this stage. As cells spread out, they are less likely to cluster within a single ST spot footprint.

Across all stages, the cross-validation reveals a fundamental spatial disconnect. Whether due to signal sparsity (4.5 PCW), systematic disagreement (6.5 PCW), or cell dispersal and static reference issues (9.5 PCW), the ST deconvolution scores and ISS cell counts do not reliably map to the same physical locations.

6 Discussion

In this study, we integrated scRNA-seq, spatial transcriptomics (ST), and in situ sequencing (ISS) datasets to investigate the developmental relationship between cardiac neural crest-like (cNCC-like) and Schwann progenitor-like (SPC-like) populations during human heart development. By combining transcriptional profiling with spatial analyses, we observed evidence for both developmental relatedness and progressive functional specialization between these populations. Our scRNA-seq analysis demonstrated that the original mixed neural crest and Schwann progenitor cluster could be separated into two transcriptionally distinct subpopulations using marker-based

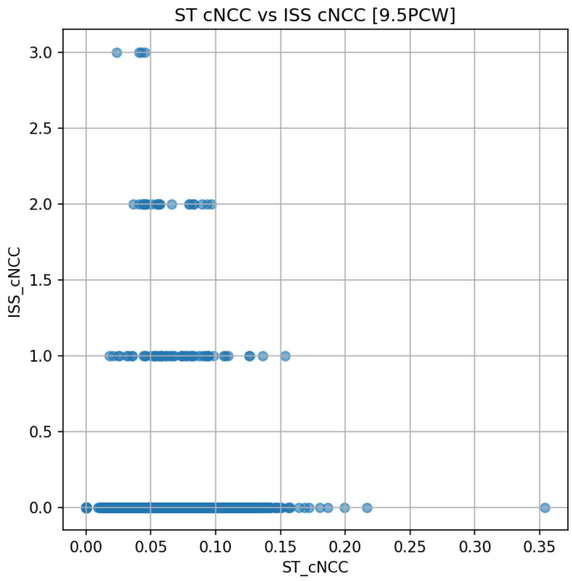

Figure 18: PCW 6.5 cNCC concordance

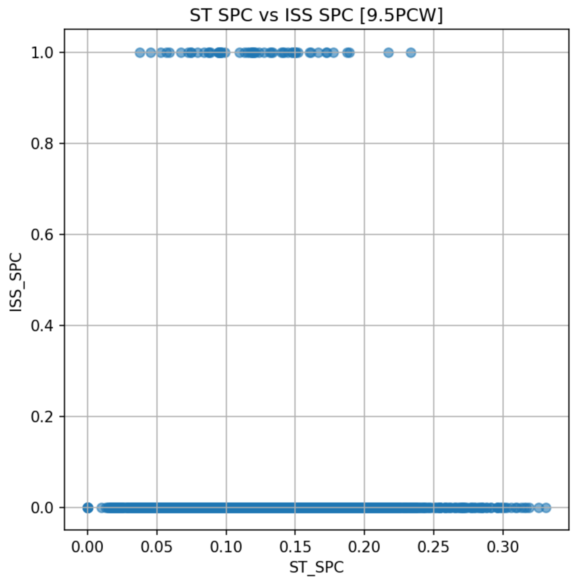

Figure 19: PCW 9.5 SPC concordance

scoring. PCA analysis revealed clear separation between cNCC-like and SPC-like cells, indicating that these groups represent biologically meaningful molecular states rather than random transcriptional variability. Marker validation further supported this interpretation, with cNCC-like cells enriched for genes associated with neural crest identity and developmental plasticity, including STMN2 and ISL1, while SPC-like cells showed elevated expression of glial and Schwann lineage markers such as ALDH1A1, ERBB3, S100B, and MPZ. These findings suggest that although both populations likely originate from neural crest-derived developmental programs, they occupy distinct transcriptional states during heart development. The differential expression analysis further strengthened this distinction. Genes associated with progenitor-like and migratory neural crest states were preferentially enriched in cNCC-like cells, whereas SPC-like cells exhibited stronger expression of genes linked to glial maturation and supportive cellular functions. Increased expression of VIM in SPC-like cells may reflect enhanced mesenchymal and structural characteristics associated with Schwann lineage differentiation. Together, these findings support a developmental model in which cNCC-like cells may transition toward a more specialized glial-associated phenotype while still maintaining partial transcriptional continuity. Spatial transcriptomic analyses provided additional insight into the temporal and anatomical organization of these populations. Module scoring across developmental stages suggested that SPC-associated signatures became more prominent during intermediate developmental windows, consistent with progressive maturation of peripheral glial populations during cardiac innervation. Meanwhile, cNCC-associated signatures appeared more broadly associated with earlier developmental processes related to migration and tissue patterning. The preservation of spatial context through ST analysis therefore helped connect transcriptional states to developmental timing within the human heart. The ISS-based spatial neighborhood analysis further demonstrated that cNCC and SPC populations were not randomly distributed throughout the tissue. Instead, cells preferentially clustered with neighboring cells of the same predicted subtype, suggesting increasing spatial compartmentalization during development. This observation supports the idea that developmental specialization is accompanied not only by transcriptional divergence but also by organization into distinct microenvironments within the developing heart. Such spatial segregation may reflect differences in local signaling environments, migratory behavior, or functional roles during cardiac morphogenesis and innervation. Several limitations should also be considered. First, the subtype classification relied on marker-based scoring using a limited number of lineage-associated genes. Although the selected markers are supported by prior literature, transitional or intermediate cellular states may not be fully captured by this approach. Second, spatial transcriptomics data have limited spatial resolution because each spot may contain multiple cells, potentially obscuring fine-scale cellular heterogeneity. While ISS analysis partially addressed this limitation, segmentation accuracy and transcript sparsity remain technical challenges for single-cell spatial inference. In addition, the developmental timepoints analyzed represent snapshots rather than continuous lineage tracing, limiting our ability to directly infer cellular transitions over time. Future studies incorporating trajectory inference, lineage tracing approaches, or higher-resolution spatial technologies could further clarify the developmental relationship between these populations. Overall, our findings support a model in which cNCC-like and SPC-like populations are developmentally related yet progressively diverge into spatially and transcriptionally specialized cell states during human heart development. The integration of scRNA-seq, ST, and ISS data provides a multidimensional framework for studying transient developmental populations and demonstrates how combining molecular and spatial information can improve understanding of complex developmental processes.

7 Conclusions

This study investigated the molecular and spatial relationship between cardiac neural crest-like and Schwann progenitor-like populations during human heart development using integrated scRNA-seq, spatial transcriptomics, and in situ sequencing datasets. Our analyses identified two transcriptionally distinct but biologically related cell populations within the developing heart. cNCC-like cells were enriched for neural crest-associated and progenitor-related markers, whereas SPC-like cells displayed stronger glial and Schwann lineage signatures. Spatial and temporal analyses further demonstrated that these populations exhibit distinct developmental organization patterns. SPC-like signatures became increasingly enriched during intermediate developmental stages, while neighborhood enrichment analysis showed preferential spatial clustering within the same predicted cell types. These findings suggest that although cNCC-like and SPC-like populations may share developmental origins, they progressively acquire distinct transcriptional identities and spatial niches during cardiac development. Together, our results support a model of partial developmental continuity accompanied by increasing functional and spatial specialization. More broadly, this work highlights the value of integrating single-cell and spatial transcriptomic technologies to study dynamic developmental processes in complex tissues such as the human heart.

8 Data and Code Availability

- **GitHub Repository:** https://github.com/Yongchean-Chhy/Spatial-and-temporal-analysis-of-cardiac-neural-c git
- **Spatial Transcriptomics Dataset (Version 3):** https://data.mendeley.com/datasets/compare/mbvhhf8m62?old=2&new=3
- **In Situ Sequencing (ISS) Dataset:** https://figshare.com/articles/dataset/ISS_data_in_A_spatiotemporal_organ-wide_gene_expression_and_cell_atlas_of_the_developing_human_heart_/10058048/1

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